

PERFECT LABELS FOR FREE? EXACT ANSWERS TEACH SHORTCUTS AND ERASE COMPUTATION; EXACT TRACES TEACH STEPS SEEN OFTEN ENOUGH

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Paper under double-blind review

ABSTRACT

Exact verifiers (primality tests, SAT solvers, unit tests) label data without noise and supply the rewards of verifiable-reward post-training. When every label is exact, a difference between two training regimes on the same inputs is a difference in the supervision format. We fine-tune Qwen2.5 models (0.5B to 7B; main size 1.5B) with LoRA on exactly labeled decision tasks and compare supervision by the answer alone with a deterministic procedure trace followed by the answer. At 1.5B, answer-only training learns what surface probes learn: on 4-digit primality its answers follow the rule “the last digit is 1, 3, 7 or 9” on 99.2–100.0% of test items, and on divisibility by 7 or 13 it stays at chance at every training size. Where the base model already computes the answer, answer-only fine-tuning erases the computation: at 3B, divisibility by 7 falls from 94.2% zero-shot to 52.3%, with the same drop at 7B, while fine-tuning on the model’s own verifier-filtered traces keeps it (96.9%). Traces teach the procedure within a task once each step is supervised often enough, but they are not uniformly better: at 1.5B they hurt on primality (55.8% vs. 82.1%), answers match or beat them on argument validity, and at 3B the long-division trace needs more examples than at 1.5B to pass the zero-shot model (75.2% at $n = 180$, 97.9% at $n = 270$ in one seed). Timestamped predictions logged during the study held within tasks and failed where step difficulty changed across tasks, and exact audits show trace-trained models writing unfaithful steps behind right answers. The results suggest that verified labels do not by themselves determine what a model learns; the supervision format and the base model’s own computation matter as well.

1 INTRODUCTION

Verifiable-reward post-training rests on programs that decide correctness exactly: answer checkers for mathematics, unit tests for code, solvers for logic (Lambert et al., 2024; DeepSeek-AI et al., 2025). Such verifiers are often described as a source of perfect labels for free, and a verified label is never wrong. Whether a model trained on those labels learns the computation the verifier performs, or a cheaper function that agrees with it on the training distribution, is a separate question, usually confounded with label noise, task difficulty and supervision format.

With exact labels the first confound is gone. We choose small decision tasks whose structure is known in advance (which surface rules approximate the label, which steps the natural procedure takes), so that we can tell which function the model learned. The point is not to teach a language model what a ten-line program computes, but to see what exact supervision of a given format teaches.

Answer-only supervision (arm A) teaches a shortcut. On 4-digit primality A’s answers coincide with “the last digit is 1, 3, 7 or 9” on 99.2–100.0% of test items, and where no surface rule exists (divisibility by 7 or 13) A stays at chance at every training size we ran. At 3B and 7B, where the zero-shot model already divides correctly, answer-only fine-tuning removes that computation (§5). *Trace* supervision (arm B: a deterministic procedure trace, then the answer) teaches the procedure once its steps are supervised often enough (Figure 1a): on divisibility by 7, accuracy rises from 50.0%

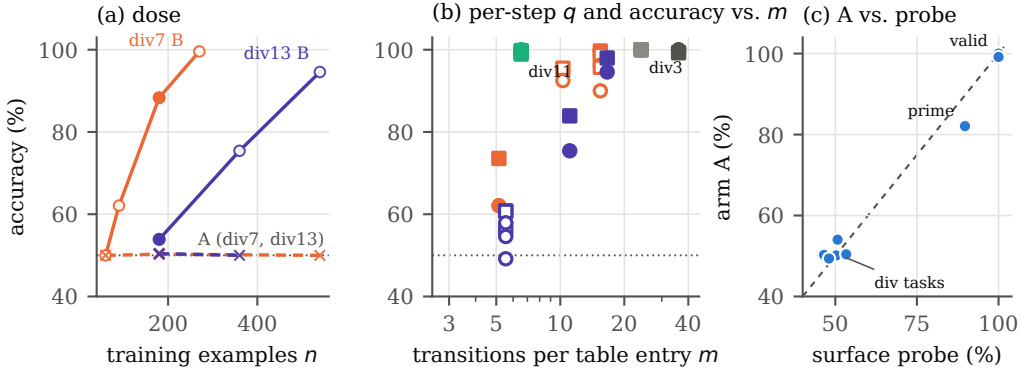


Figure 1: (a) Accuracy against training examples n for div7 and div13 at 1.5B: trace arm B (solid) and answer arm A (dashed; chance at every n); filled markers are means over two or three seeds, hollow markers single seeds. (b) Arm B per-step accuracy q (squares; probability that a step is right given that the previous one was) and answer accuracy (circles) against $m = kn/(10d)$, the supervised transitions per (remainder, digit) table entry, known before training; orange div7 (4 and 6 digits), violet div13, green div11, gray div3. Hollow: cells observed before S7 was registered; filled: cells predicted in advance by S7 or S9. (c) Answer arm A against the surface probe fitted on the same training items (one-hot features where engineered features encode the answer); dashed line: equality.

at $n = 60$ training examples to 99.6% at 270. It is not a free improvement: at 1.5B the primality trace hurts, on argument validity answers do as well, and at 3B and $n = 180$ the long-division trace underperforms the zero-shot model.

Contributions.

- Timestamped dose predictions for trace supervision that held within tasks and failed where step difficulty changed across tasks, exposing step difficulty as a confound of supervision frequency (§4.2).
- Evidence that answer-only fine-tuning on exact labels teaches the surface probe’s function (at 1.5B, and on primality at 3B and 7B; div3 at 3B, learned from answers while the probe stays at chance, is the exception) and erases the base model’s own computation at 3B and 7B (and lowers div7 below base at 1.5B), while fine-tuning on its own verifier-filtered traces retains it at 3B (two seeds) and 7B (one seed) (§4.1, §5).
- Exact-verifier audits of the traces themselves, showing that trace-trained arms also learn shortcuts behind unfaithful intermediate steps (§4.2, §5); all failed predictions are reported (Table 4).

2 RELATED WORK

Intermediate steps as supervision. Scratchpads (Nye et al., 2021) and chain-of-thought (CoT) prompting (Wei et al., 2022) improve multi-step computation, mainly on mathematical and symbolic tasks (Sprague et al., 2025). Data format changes the sample efficiency of arithmetic learning (Lee et al., 2024), fine-tuned models can learn arithmetic with decomposed steps (Liu & Low, 2023), and GCD predictions follow rules built from divisors of the number base (Charton, 2024), the kind of surface rule we probe for. Errors compound across steps (Dziri et al., 2023); filler tokens can replace a chain of thought on some tasks (Pfau et al., 2024), and our scrambled control keeps the tokens but removes their order. Reasoning helps when training data are local, so that intermediate variables are seen together often (Prystawski et al., 2023); our dose results are a within-task instance of that locality account, and they show where it stops predicting across tasks. Distilling rationales into small models (Hsieh et al., 2023; Ho et al., 2023; Magister et al., 2023) trains on traces before answers; placing explanations after the answer (Lampinen et al., 2022) and internalizing steps (Deng et al., 2023; 2024) are the alternatives our answer-first arm B’ probes, and on div7 it does not transfer the

Table 1: Tasks. k : steps in the trace (prime: for primes; a composite’s trace stops at its smallest factor). Tokens: mean whitespace tokens per trace. m : supervised transitions per (remainder, digit) table entry at $n = 180$.

task	trace procedure	k	tokens	m
prime	trial division by the primes $\leq \lfloor \sqrt{n} \rfloor$	11–25	71.5	–
div d , 4 digits	long division, $r \leftarrow (10r + x_t) \bmod d$	4	58.0	36.0 ($d=2$) ... 5.5 ($d=13$)
div7, 6 digits	same, $d = 7$	6	76.0	15.4
valid	Z3 countermodel or unsatisfiable core	certificate	37.0	–

procedure. Transformers can also learn multi-hop composition implicitly after long training (Wang et al., 2024); our answer arm, at a few hundred examples, does not.

Theory. Parity-like targets give gradient-based learners almost no signal from end labels (Shalev-Shwartz et al., 2017). Sub-task decomposition makes otherwise unlearnable tasks learnable (Wies et al., 2023); CoT raises the expressivity of bounded-depth transformers (Feng et al., 2023; Li et al., 2024; Merrill & Sabharwal, 2024), next-token predictors trained on CoT data are universal learners (Malach, 2024), and CoT induces sparse, sample-efficient dependence (Wen et al., 2025). Transformers can represent automata through shallow shortcuts (Liu et al., 2023). The globality degree of Abbe et al. (2024) bounds what transformers learn efficiently; teacher-forced intermediate parities make parity learnable (Kim & Suzuki, 2025). Grokking studies how modular tables are learned from a fraction of their entries (Power et al., 2022). Section 6 uses these results as intuition only.

Traces and verifiers in post-training. Transformers trained from scratch on verifiable traces depend less on trace semantics than usually assumed (Valmeekam et al., 2026), and CoT explanations can be unfaithful to the computation behind the answer (Turpin et al., 2023); our exact audits measure this in fine-tuned traces. Process supervision rewards steps rather than outcomes (Lightman et al., 2024); program-generated logic corpora improve reasoning (Morishita et al., 2024). STaR (Zelikman et al., 2022) and rejection-sampling fine-tuning (Yuan et al., 2023) train on self-generated rationales that reach verified answers, as our arm S does; RL with verifiable rewards drives recent reasoning models (Lambert et al., 2024; DeepSeek-AI et al., 2025). Fine-tuning on model-generated positives can amplify spurious correlations (Setlur et al., 2024); SFT and RL generalize differently (Chu et al., 2025); RLVR mostly reweights what the base model already samples (Yue et al., 2025). Our answer arm is an instance of shortcut learning (Geirhos et al., 2020); exact labels do not prevent it.

3 SETUP

Tasks. Each task has an exact decision procedure, so labels are correct by construction; arms see the same prompts and labels and differ only in the completion (Tables 1 and 2). **prime**: is a 4-digit n prime? The trace performs trial division by the primes up to $\lfloor \sqrt{n} \rfloor$, one “ $n \bmod p = r$.” step per prime, stopping at the first divisor. **div d** : is n divisible by d ? The trace performs long division, $r \leftarrow (10r + x_t) \bmod d$ for each digit x_t , for $d \in \{2, 3, 7, 11, 13\}$ with 4 digits and $d = 7$ with 6. **valid**: does a propositional conclusion follow from its premises (Z3 (de Moura & Bjørner, 2008))? Arguments instantiate 12 forms shared between training and test, so the task is surface-learnable by design (Appendix C). Test sets have 240 items (120 per class), are problem-disjoint from training and identical across arms; training sets are balanced samples of size n drawn with the run’s seed.

Training and evaluation. LoRA (Hu et al., 2022) with $r = 8$, $\alpha = 16$, dropout 0.05 on the attention projections of Qwen2.5-Instruct (Qwen et al., 2024) (0.5B, 1.5B, 3B, 7B); AdamW, learning rate 2×10^{-4} , batch size 1, 3 epochs, bf16; loss on completion tokens only; one configuration for every arm, not tuned per arm. Evaluation is greedy generation with one answer extractor for every arm; an unparsed or truncated answer counts as wrong. Details and generation caps: Appendix B.

Statistics and timestamped predictions. We lead with the difference of two arms and its 95% bootstrap interval, resampling items and seeds jointly, and give the number of seeds; per-seed exact

Table 2: Arms (completions trained on; the prompt is shared) and the labels of the tests. H1–H4: endpoints of a first preregistration (prior grid). S1–S10: timestamped predictions of the second file (Table 4).

arm	completion
base	none (zero-shot); also a zero-shot CoT prompt and 4 trace demonstrations
A	“Answer: Yes” or “Answer: No”
B	the procedure trace, a newline, the answer
C	B’s traces permuted across examples and token-shuffled; the answer keeps the true label
D (S8)	a fluent, correct trace of a <i>different</i> number with the same label
B’	the answer first, then the trace
S (S10)	the base model’s own sampled solution, kept when the exact gate accepts its final answer (STaR-style)
A-full	answers on the whole training pool

Table 3: Accuracy (%) at $n = 180$, Qwen2.5-1.5B-Instruct, 240 test items per task; mean over seeds $\pm$ SD across seeds (seeds per cell: Table 6; no SD: one seed). B–A: points, with a 95% bootstrap interval over items and seeds. max p : largest per-seed exact McNemar p (B vs. A); in parentheses where $A > B$. -: not run.

task	base	probe	A	B	C	B–A [95% CI]	max p
prime	59.6	89.7	82.1 $\pm$ 0.4	55.8 $\pm$ 3.3	52.1 $\pm$ 2.9	-26.2 [-34.9, -17.2]	(5.7×10^{-7})
div7	62.5	46.5	50.3 $\pm$ 2.1	88.3 $\pm$ 5.2	50.0 $\pm$ 0.6	+38.1 [+30.8, +44.0]	6.8×10^{-13}
valid	30.0	100.0	99.2 $\pm$ 0.8	88.3 $\pm$ 10.4	52.5 $\pm$ 2.4	-10.8 [-20.6, +0.4]	(0.50)
div3	78.3	50.7	54.0 $\pm$ 7.3	100.0 $\pm$ 0.0	–	+46.0 [+37.8, +54.4]	1.6×10^{-27}
div11	55.0	50.3	50.1 $\pm$ 0.9	99.4 $\pm$ 0.5	–	+49.3 [+43.3, +55.3]	7.2×10^{-34}
div13	50.4	53.3	50.4 $\pm$ 0.8	53.9 $\pm$ 4.4	45.0	+3.5 [-4.6, +12.4]	0.86
div7 (6 digits)	52.5	48.1	49.4 $\pm$ 2.1	93.1 $\pm$ 4.4	–	+43.8 [+34.6, +52.5]	4.5×10^{-23}

McNemar tests (McNemar, 1947), Holm-corrected (Holm, 1979) across tasks within a comparison, model and seed at $\alpha = 0.05$, are secondary, because they count test items, not training runs. A first preregistration, written before a prior grid of 36 runs according to the history of the repository it was written in, fixed H1–H4. A second file, committed at 20:29:47 on 2026-09-25 before any of its runs, fixes S1–S6; its append-only decision log adds S7–S10, each committed before any cell it governs had started (Appendix E). These are timestamped predictions logged during an adaptive study, not one up-front plan: S7–S10 were proposed by the AI assistant after earlier results were in. The tests at risk were S1, S2, S4, S5 and S7–S10; S3 and S6 were effectively settled before registration. The surface probe and every analysis labelled post hoc belong to neither file.

4 RESULTS

4.1 EXACT ANSWER LABELS TEACH THE SURFACE

The primality shortcut (descriptive, post hoc). A reaches 82.1% on prime (seeds 0–2), and its answers coincide with “the last digit is 1, 3, 7 or 9” on 99.2–100.0% of test items; that rule alone scores 82.5%. A classifies 99.7% of primes and 64.4% of composites correctly, and of the 42 test composites ending in 1, 3, 7 or 9 it gets 0, 0, 0 right in seeds 0–2, including 27 with a factor 3 or 7 that any sieve would catch. On a dedicated set of 120 odd composites without a factor ≤ 7 and as many primes (S6, passed, settled in advance), A reaches 49.6% and agrees with the rule on 98.8% of items. We named the rule after seeing the prior grid, so this analysis describes rather than tests.

Answers track surface probes. We fit logistic regression, a one-hidden-layer MLP and k -nearest neighbours on the same 180 training items as arm A, on 40 one-hot digit-by-position features alone or with engineered divisibility features (last digit even or 0/5; digit sum mod 3 and 9; alternating digit sum mod 11), or for arguments on atom-renamed word and character n -grams, and choose the probe by 5-fold cross-validation; on div2, div3 and div11 the engineered features contain the answer, so there the probe is one-hot only. A tracks the probe (Figure 1c): on prime it lies between

the one-hot linear probe (79.3%) and the best probe (89.7%), on div2 both are at ceiling (100.0% and 100.0%) and on every other divisibility task both are at chance, and on valid both are near ceiling.

More answers do not buy the program. On prime, A reaches 59.6%, 82.1% and 75.0% at $n = 60, 180$ and 540 , and 80.8% on the full pool. On div7 it stays at chance at every n (50.0%, 50.3%, 50.0%) and on div13 likewise (50.4% and 50.0% at 180 and 360), and at $n = 540$ (div7) and $n = 360$ (div13) A collapses to answering “No” under the untuned configuration (Limitations); on div7 it falls below the zero-shot base (62.5%; A—base -12.2 [-19.2, -4.7] points over 3 seeds; significant after Holm correction in 2 of 3 seeds, $p_{\text{Holm}} = 0.19, 0.018, 1.9 \times 10^{-4}$). With labels permuted across items, the earlier study’s answer arm sat at 48.5–52.5% on all seven of its tasks (Appendix A): correct labels drive A, and on primality what they buy is the rule.

4.2 EXACT TRACES TEACH STEPS SEEN OFTEN ENOUGH

Divisibility by 7. At $n = 180$, over three seeds, B reaches 88.3% (92.5, 90.0, 82.5) against A’s 50.3% (B—A +38.1 [+30.8, +44.0]), and C stays at 50.0%. Seeds 1 and 2 are the preregistered replication (S1 passed); a deterministic re-execution of seed 0 gave 92.5%. The gain holds on both classes (82.2% on multiples, 94.4% on non-multiples).

Dose. With the trace format fixed, accuracy rises with the number of training traces while A stays flat (Figure 1a, where hollow markers are single seeds). Div7 B goes from 50.0% at $n = 60$ to 62.1% at 90 (per-step accuracy $q = 0.736$), 88.3% at 180 and 99.6% at 270 ($q = 0.997$). Div13 B, with the same 4 steps, goes from 53.9% at 180 (seed-0 $q = 0.566$) to 75.4% at 360 and 94.6% at 540 ($q = 0.980$).¹ At $n = 180$, div13’s failure is visible in its seed-0 generations: 2.1% of traces are correct, the first error comes at step 2 in 188 of 240 items, and 179 of 235 wrong traces end at $r = 0$, so this seed answers “yes” 76% of the time (the bias is seed-dependent; Table 6). In div7 generations (seed 0, the only seed with saved generations at this n), digits and carried remainders are copied without error (100.0%, 100.0%); every error is a wrong lookup of $(10r + x) \bmod 7$.

Timestamped dose predictions (S7, S9): held within tasks, failed across tasks. S7 predicted that q follows $m = kn/(10d)$ in five cells and passes if four hold. The three within-task cells hold: (a) div13 at $n = 540$ ($m = 16.6$), $q = 0.980$ and accuracy 94.6%; (b) div7 at $n = 270$ ($m = 15.4$), $q = 0.997$; (c) div7 at $n = 90$ ($m = 5.1$), $q = 0.736$ and accuracy 62.1%. Cell (e), div3 and div2, holds but at ceiling ($q = 1.000, 0.993$), so it tests little. Both informative cross-task tests fail ((e) held only at ceiling). (d) Div11 at $m = 6.5$ reached $q = 0.994$, above the predicted interval: $10 \equiv -1 \pmod{11}$ makes its step a subtraction. S9 (failed): div13 at $n = 360$ ($m = 11.1$) reached 75.4% against the required 80%, while div7 at the similar $m = 10.3$ reached 88.3%; the div13 step, $r' = (x - 3r) \bmod 13$, is harder. S7 thus passed by its 4-of-5 rule (4/5), but the supported reading is narrower: within a task more supervised transitions raise q ; across tasks step difficulty shifts the curve, and m does not transfer across tasks or sizes (post hoc logistic fits of q on $\ln m$: $R^2 = 0.41$ over 17 cells at 1.5B, 0.34 over 13 at 3B). S2 passed at 1.5B: B—A ≥ 15 points in both seeds for div3 (+50.0, +37.5) and div11 (+48.3, +50.0), not for div13 (+5.0, -1.2).

Length (S5): failed as preregistered. The prediction for 6-digit div7 was $B \approx q^6 = 75.8\%$, with $q = 0.955$ measured on the seed-0 4-digit generations. The primary cell, B trained and tested on 6-digit traces, reached 96.2% and 90.0% in seeds 0 and 1, outside the 10-point tolerance: S5 failed. Its own seed-0 q is 0.988: at $k = 6$ the same n gives more transitions per entry ($m = 15.4$). The secondary cell, the 4-digit model applied to 6-digit inputs, gives 77.1%, within the tolerance, but its own q is 0.906 and only 48.3% of its traces are fully correct, so the match is not evidence for compounding (post hoc forms: Appendix F).

Primality: the trace hurts at 1.5B. B reaches 55.8% against A’s 82.1% (seeds 0–2; B—A -26.2 [-34.9, -17.2]), and B vs. C is not significant ($p = 0.22, 0.33$). B classifies 19.7% of primes and 91.9% of composites correctly. Class and trace length are confounded here: every prime needs the full sequence of trial divisions, and accuracy falls with the true number of divisions (100, 89, 71,

¹Sensitivity check, not the preregistered measure: excluding the near-trivial first step ($r_1 = x_1 \bmod d$), seed-0 q is 0.237 instead of 0.566 for div13 at $n = 180$ and 0.602 instead of 0.736 for div7 at $n = 90$.

Table 4: Timestamped predictions (S1–S6 before any of their runs; S7–S10 appended during the study, each before its cells) and their status. 3B and 7B were not the preregistered model; their columns are secondary; $[k]$: number of seeds. –: not run.

	prediction	result (1.5B)	status	3B	7B
S1	div7 B $\geq$ 80, B–A $\geq$ 20 pp (seeds 1, 2)	B 90.0/82.5, A 50.8/47.9	passed		
S2	traces help $\geq$ 2 of div3/11/13 (2 seeds)	B–A div3 +50.0/+37.5, +48.3/+50.0, div13 +5.0/–1.2	passed	failed (div3 +0.8)	div3 –
S3	div2 A $\geq$ 95	100.0	passed (settled)		
S4	div7 B > A at 3B (2 seeds)	(3B) A 54.6/50.0, B 68.3/82.1	passed		
S5	6-digit div7 B within 10 pp of $q^6 = 75.8$	96.2/90.0	failed	failed	–
S6	prime A on hard set < 60, follows rule	49.6, rule agreement 98.8	passed (settled)	passed	–
S7	q follows m ; $\geq$ 4 of 5 cells	(a)–(c) hold, (e) at ceiling, (d) fails	passed (4/5)	post hoc only	
S8	arm D $\leq$ 60 (2 seeds)	50.0/50.4	passed	D 50.0 [1]	D 47.9 [1]
S9	div13, $n = 360$: B $\geq$ 80, A $\leq$ 60	B 75.4, A 50.0	failed	passed (83.3)	–
S10	arm S div7 $\leq$ 65 (2 seeds); valid $\geq$ 90	div7 77.9 (seed 0)	failed	contradicted (div7 97.5/96.2)	div7 82.1 [1]

26 and 21% for 1, 2–3, 4–8, 9–16 and 17 or more). Scrambled C does not separate the two: seed 1 shows B’s split, seed 0 the opposite (99.2% of primes, 0.8% of composites). At $n = 540$, B reaches 68.8% against A’s 75.0% (one seed). A trial-division step is a table indexed by (n, p) with more than 10^9 entries, so $m \ll 1$ at these n . At 3B the trace arm scores higher (85.0%) without executing its trace: only 23% of its “ $n \bmod p = r$ ” claims are correct, seed 0 agrees with “no factor ≤ 5 , so prime” on 100.0% of items, and on the hard set it scores 50.4% and 53.8%: a shortcut written in the form of a trace (cf. Turpin et al., 2023).

Validity and cost. On valid, answers match or beat traces (A 99.2%, B 88.3%, C 52.5%; B–A -10.8 $[-20.6, +0.4]$; per seed $p = 0.50$, 7.1×10^{-15} and 4.7×10^{-10} for seeds 0–2). B’s completions contain 14–47 times as many tokens as A’s (Qwen tokenizer). Of the first preregistration’s endpoints, H1 held on div7 only, H2 on div7 and valid, H3 failed where answers work (it was not tested on div7), and H4 held at $n = 180$ but not at $n = 60$.

4.3 CONTROLS (DIV7, 1.5B)

The trace must be the input’s own computation. D, a fluent and correct trace of a different number with the right label, gives 50.0% and 50.4% (S8 passed): a correct trace of another number does not help. Scrambled C stays at 50.0%. C controls for token count but not for interference: it also fails where A succeeds (valid 52.5% vs. 99.2%, prime 52.1% vs. 82.1%), so C alone does not show that the trace’s information is what helps.

The trace must come before the answer. B’, trained with the trace after the answer, gives 48.3% and 50.8%, like A, although it still writes a trace after answering with $q = 0.841$.

Prompting the 1.5B base model does not replace training. A zero-shot CoT prompt gives 67.1% on div7, 62.9% on prime and 66.7% on valid; 4 trace demonstrations give 47.9%, 39.2% and 68.8%. No prompt approaches B on div7.

Arm S at 1.5B (S10). Sampling from the base model’s CoT prompt and keeping completions the gate accepts covers 159/180 training items (keep rate 56.9% of samples). S10 predicted div7 accuracy $\leq 65\%$ in seeds 0 and 1, fewer than half of kept traces with all remainders correct, and valid $\geq 90\%$. Result: arm S reaches 77.9% on div7 (seed 0), against 67.1% for the base model’s CoT prompt; S10: failed. The trace audit depends on its parser: the worker’s audit finds a checkable claim in 51.6% of kept traces and counts 36.5% of all kept traces as fully correct, while a post hoc re-audit that also reads \div, the division sign and truncated decimals parses 89.9% and finds 69.9% of the parsed traces correct (62.9% of all kept), so the audit half of S10 holds under the first parser and fails under the second.

Table 5: Scale: zero-shot base / answers A / trace B at $n = 180$, accuracy in %, mean over seeds (seeds per cell: Table 6). –: not run. [†]Lower bound: 110 of 240 generations hit the cap.

task	0.5B	1.5B	3B	7B
div7	– / 50.4 / 79.2	62.5 / 50.3 / 88.3	94.2 / 52.3 / 75.2	82.5 / 51.0 / 88.5
div13	– / – / –	50.4 / 50.4 / 53.9	92.1 / 49.2 / 56.0	80.0 / 46.7 / 63.5
div3	– / – / –	78.3 / 54.0 / 100.0	– / 99.2 / 100.0	–
div11	–	55.0 / 50.1 / 99.4	– / 53.1 / 99.8	– / 55.4 / 98.8
div7, 6 digits	–	52.5 / 49.4 / 93.1	– / 49.2 / 82.5	–
prime	– / 68.3 / 57.5	59.6 / 82.1 / 55.8	47.1 [†] / 82.3 / 85.0	72.1 / 66.9 / 84.6
valid	– / – / –	30.0 / 99.2 / 88.3	85.8 / 100.0 / 95.6	89.6 / 100.0 / 58.3

5 SCALE

Table 5 repeats the core cells at 0.5B, 3B and 7B. Three results replicate. Where no surface rule exists (div7, div13, div11, 6-digit div7), answer-only training stays at chance at every size. The primality shortcut persists: at 3B, A reaches 82.3% and agrees with the last-digit rule; at 7B one seed does the same (82.5%) while the other collapses to answering “No” on 237 of 240 items (51.2%), so the mean 66.9% averages two different failures, against B’s 84.6% (89.2, 80.0). Trace supervision beats answers on div7 at every size (S4 passed; B–A +22.9 [+9.6, +37.9] at 3B and +37.5 [+30.8, +44.2] at 7B, two seeds each), and the div7 controls replicate: D gives 50.0% at 3B and 47.9% at 7B, B’ 50.8% and 53.3% (seeds per cell: Table 6). At 7B, div11 behaves as at 1.5B (A 55.4%, B 98.8%).

Some cells move with size. At 3B, div3 becomes learnable from answers alone (A 99.2%, against 54.0% at 1.5B), so S2 fails at 3B. The S9 threshold is met at 3B (B 83.3%, A 50.0%). At 7B the trace arm collapses on valid (seed 0: 58.3%): all 100 of its 100 errors are invalid arguments answered “yes”, 99 of them after a trace asserting a contradiction: the certificate’s form without its check.

Answer-only fine-tuning erases the base model’s computation. The 3B and 7B zero-shot models already decide divisibility: 94.2% and 92.1% on div7 and div13 at 3B, 82.5% and 80.0% at 7B. The 3B base model writes one direct division line on 100.0% of its 240 div7 answers (“2740 ÷ 7 = 391.428571 . . . , not an integer”). Answer-only fine-tuning removes it: div7 A falls to 52.3% at 3B (A–base -41.9 [-48.8, -34.6]) and 51.0% at 7B (-31.5 [-39.2, -23.5]), and div13 A to 49.2% and 46.7% (means over two seeds at 3B and two seeds at 7B; Table 6). Fine-tuning on the model’s own verifier-filtered solutions (arm S) keeps the computation: at 3B, S reaches 96.9% on div7 (97.5, 96.2), above the base in seed 0 (discordant pairs 8:0, $p = 0.0078$), and its kept traces are right (the post hoc re-audit parses 97.7% and finds 96.4% fully correct; the worker’s narrower audit parsed 15.0%). At 7B, S gives 82.1% in seed 0, level with the base (discordant 13:14). The foreign long-division trace depends on size: at 3B it stays below the base on div7 (75.2%; B–base -19.0 [-30.4, -8.3]) and div13 (56.0%); at 7B it matches the base on div7 (88.5%; +6.0 [-0.8, +12.5]) but not on div13 (63.5%); at 1.5B, whose base scores 62.5%, it exceeds the base. More traces close the 3B gap, a within-task replication of the dose result (not preregistered at 3B): div7 B rises from 60.0% at $n = 90$ to 97.9% at 270 (seed 0, one seed), above the zero-shot model, and div13 B reaches 99.6% at $n = 540$ while A stays at 50.0%.

6 AN AUTOMATON VIEW (INTUITION)

Remark 1 (intuition, not a theorem). Let $x_1 \cdots x_k$ be the digits of n and $\delta_d(r, a) = (10r + a) \bmod d$. Then $d \mid n$ iff $\delta_d^*(0, x_1 \cdots x_k) = 0$: divisibility by d is recognized by a d -state automaton reading digits, whose states the arm-B trace writes. Under teacher forcing each supervised state is a function of two tokens already in the context, so n traces are kn samples of one table with $10d$ entries, $m = kn/(10d)$ per entry. Answer-only supervision gives n samples of the k -fold composition, and for d coprime to 10 the label is nearly independent of every proper subset of the digits: for $d = 7$ and $k = 4$, fixing any three digits leaves $\Pr(7 \mid n)$ between 0.10 and 0.22 (marginal 0.14).

The remark resembles the globality barrier of Abbe et al. (2024) and the teacher-forced CoT of Kim & Suzuki (2025), but those concern growing input length and do not apply literally to fixed 4-digit inputs. It counts how often each transition is supervised, not how hard it is: for $d = 3$ and $d = 11$

the transition is an addition or subtraction mod d , for $d = 7$ and $d = 13$ it involves a multiplication, and the results follow difficulty as well as m .

7 DISCUSSION AND LIMITATIONS

What exact labels guarantee. An exact verifier guarantees that no training label is wrong, not that the labels identify the verifier’s function among those a model can fit. The rule A learned on primality calls every composite ending in 1, 3, 7 or 9 prime; by construction that includes all 996 of the 7,939 4-digit composites without a factor ≤ 7 . Shuffled labels show that the gain comes from label correctness but cannot tell the verifier’s function from a surface rule; inputs on which the rule is wrong can.

Answer-only fine-tuning erases the base model’s test-time computation. Labels are exact in every arm, so the effect comes from the supervision format and the information it exposes; we ran no noisy-label arm. At 3B and 7B, answer supervision replaced a working computation with a guess, in both seeds; at 1.5B, A falls below the zero-shot base on div7 in the same direction (significant after Holm correction in 2 of 3 seeds). Fine-tuning on the model’s own verifier-filtered solutions kept the computation (3B, two seeds; 7B, one seed). A pipeline that fine-tunes a capable model on verified answers alone should compare against the model’s own zero-shot accuracy, not only against chance.

Supervision design: hypotheses for verifiable-reward pipelines. Our results suggest, without establishing beyond these tasks: (1) fit a cheap surface probe on the verified data before training; where it is accurate, answer supervision may teach its function, so evaluate where it is wrong. (2) A correct trace is not enough: budget how often each step type is supervised, expect harder steps to need more, and check per-step accuracy after a small run. (3) The trace should be the input’s own computation and precede the answer (tested on div7 only). (4) Check the base model first: where its own steps are right, its verifier-filtered traces preserved them in our 3B (two seeds) and 7B (one seed) runs, and a foreign trace format underperformed at 3B at $n = 180$. (5) Do not read a null from one training size: our 60-example null on div7 was the bottom of a dose curve.

RLVR: a hypothesis, not a result. With a binary verifier the RLVR reward depends on the final answer only. We hypothesize, without testing, that RLVR then behaves like answer supervision with on-policy sampling: on a surface-learnable task a surface rule already earns most of the reward (the last-digit rule is right on 82.5% of our prime test items), and on div7 the policy would find the procedure only if its own samples sometimes execute it (Yue et al., 2025). We ran no RL; arm S is the nearest experiment.

Limitations. One model family, with 1.5B as the main size, and one hyperparameter configuration shared by all arms (LoRA $r = 8$ on the attention projections only, learning rate 2×10^{-4} , batch size 1, 3 epochs), not tuned per arm; answer-only failure may reflect sample inefficiency under this configuration rather than impossibility, and A was tested only up to $n = 540$. B sees 14–47 times more supervised tokens than A, and the token-matched comparison (H3) was not run on div7. Binary tasks with 240 test items and one to three seeds per cell; most dose points have one seed (Figure 1a). One trace format per task. Some base-model cells are lower bounds because generations hit the cap (Appendix B), and A collapses to a constant answer in some seeds (Table 6). S5 and S9 failed; S7 passed by its 4-of-5 rule, but m is confounded with step difficulty; S7–S10 were added during the study; S3 and S6 were settled before registration; S10: failed. The trace audits depend on the parser (§4.3). The shuffled-label control comes from an earlier study; the probe was defined after the prior grid; the logic task shares its forms between training and test.

AI USE STATEMENT

We used an AI assistant (Claude, Anthropic) to write and refactor experiment code, to run the experiments and log their outputs under our direction, to compute the statistics, to draft and edit the manuscript, and to check every bibliography entry against arXiv, OpenReview, proceedings and DOI records. The research question, the arms and decision rules S1–S6 were reviewed and approved by

the authors before the runs they govern. Hypotheses S7–S10 were proposed by the assistant while the experiments ran, committed to the repository before any of their cells started (commit times in Appendix E), and reviewed by the authors afterwards; the interpretation is the authors’. We reviewed all AI-assisted code, text and numbers, verified reported numbers against the result files, and take responsibility for the content of this paper.

REPRODUCIBILITY STATEMENT

Code and data will be released: dataset generators with fixed seeds, the training and evaluation worker, queue runners, analysis scripts, both preregistration files with an anonymized timestamp manifest of their commits, and every run’s result record with per-item correctness (local paths removed), from which each paired test can be recomputed; an anonymized copy is in the supplementary material. Section 3 and Appendices B and D give prompts, trace formats and hyperparameters. Every number in the text is a macro regenerated from the result files (`paper/exp/build_all.py`).

ETHICS STATEMENT

The study uses synthetic arithmetic and logic data only and involves no human subjects or personal data.

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A FULL RESULTS

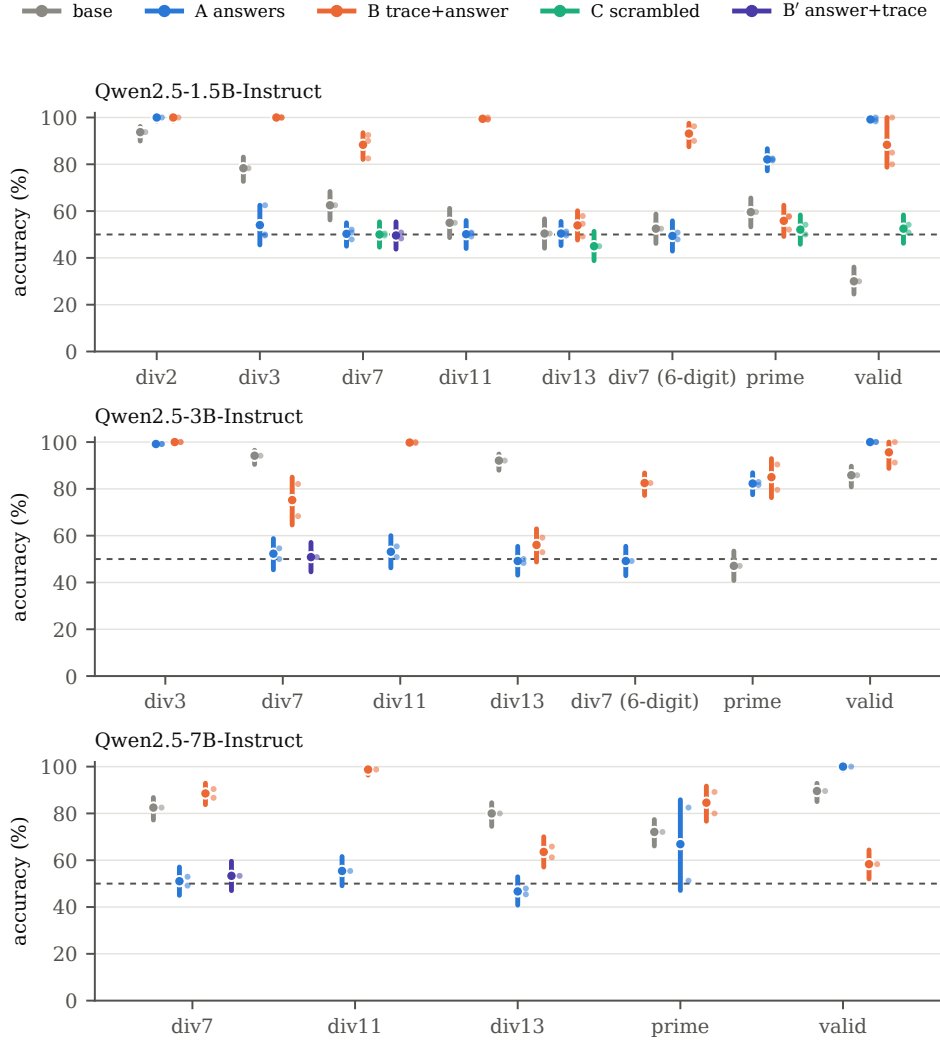


Figure 2: Accuracy by arm and task at $n = 180$; bars: 95% bootstrap interval over items and seeds where a cell has two or more seeds, Wilson interval for a single seed; small dots are single seeds; one row per model with at least three tasks (the 0.5B cells are in Table 5). Dashed line: majority class.

Table 6 lists every logged run of the arms in Table 2 and of B-ext, which uses reasoning traces sampled from a locally served 9B model and kept only when their final answer matched the verifier; only 9 traces passed, so these runs are not comparable to B and are listed for completeness. Table 8 reports the earlier answer-only study with its shuffled-label control. Per-seed exact McNemar tests (B vs. A, 1.5B): prime $p = 5.7 \times 10^{-7}$, 1.7×10^{-7} , 8×10^{-11} ($A > B$); div7 4.1×10^{-27} , 2×10^{-18} , 6.8×10^{-13} ; valid 0.50, 7.1×10^{-15} , 4.7×10^{-10} .

Table 6: All logged runs of arms base, A, B, C, D, B', S, A-full and B-ext (240 test items per cell). Cell: the test set, prefixed by the training task when they differ; [cot] and [fewshot4] mark prompting controls on the base model. Train tokens: whitespace tokens in the completions. Cap: test generations that hit the generation cap (scored wrong; – where not logged). Yes: share of test items answered Yes, from saved generations or recovered from per-item correctness; 0 or 100 marks a constant-output collapse.

model	cell	arm	n	seed	train tokens	cap	Yes (%)	acc. (%)
0.5B	prime	A	180	0	360	0	47	68.3
0.5B	prime	B	180	0	13,370	0	12	57.5
0.5B	div7	A	180	0	360	0	5	50.4
0.5B	div7	B	180	0	10,800	0	34	79.2
1.5B	prime	base	0	0	0	–	28	59.6
1.5B	prime	A	60	0	120	–	90	59.6
1.5B	prime	A	180	0	360	–	67	82.1
1.5B	prime	A	180	1	360	–	68	82.5
1.5B	prime	A	180	2	360	0	68	81.7
1.5B	prime	A	540	0	1,080	–	75	75.0
1.5B	prime	B	60	0	4,355	–	12	57.1
1.5B	prime	B	180	0	13,370	–	11	57.9
1.5B	prime	B	180	1	13,260	–	12	57.5
1.5B	prime	B	180	2	13,080	13	16	52.1
1.5B	prime	B	540	0	40,095	–	28	68.8
1.5B	prime	C	180	0	12,561	–	99	50.0
1.5B	prime	C	180	1	13,120	–	12	54.2
1.5B	prime	A-full	1320	0	2,640	–	69	80.8
1.5B	prime	B-ext	9	0	896	–	20	62.1
1.5B	prime	B-ext	9	1	974	–	12	60.0
1.5B	prime	S	180	0	18,790	1	41	67.1
1.5B	prime→prime_hard	A	180	0	0	0	99	49.6
1.5B	prime[cot]	base	0	0	0	8	28	62.9
1.5B	prime[fewshot4]	base	0	0	0	104	9	39.2
1.5B	div7	base	0	0	0	–	72	62.5
1.5B	div7	A	60	0	120	–	88	50.0
1.5B	div7	A	180	0	360	–	24	52.1
1.5B	div7	A	180	1	360	–	82	50.8
1.5B	div7	A	180	2	360	–	94	47.9
1.5B	div7	A	540	0	1,080	–	0	50.0
1.5B	div7	B	60	0	3,600	–	73	50.0
1.5B	div7	B	90	0	5,400	0	52	62.1
1.5B	div7	B	180	0	10,800	–	47	92.5
1.5B	div7	B	180	1	10,800	–	44	90.0
1.5B	div7	B	180	2	10,800	–	41	82.5
1.5B	div7	B	270	0	16,200	0	50	99.6
1.5B	div7	C	180	0	10,784	–	72	49.6
1.5B	div7	C	180	1	10,824	–	30	50.4
1.5B	div7	B'	180	0	10,800	0	48	48.3
1.5B	div7	B'	180	1	10,800	0	11	50.8
1.5B	div7	D	180	0	10,800	0	100	50.0
1.5B	div7	D	180	1	10,784	0	0	50.4
1.5B	div7	S	180	0	7,473	1	57	77.9
1.5B	div7→div7_6d	B	180	0	0	0	32	77.1
1.5B	div7[cot]	base	0	0	0	0	70	67.1
1.5B	div7[fewshot4]	base	0	0	0	0	37	47.9
1.5B	valid	base	0	0	0	–	38	30.0
1.5B	valid	A	60	0	120	–	50	97.5
1.5B	valid	A	180	0	360	–	51	99.2
1.5B	valid	A	180	1	360	–	50	100.0
1.5B	valid	A	180	2	360	0	48	98.3
1.5B	valid	A	540	0	1,080	–	50	100.0

model	cell	arm	n	seed	train tokens	cap	Yes (%)	acc. (%)
1.5B	valid	B	60	0	2,343	–	51	87.5
1.5B	valid	B	180	0	7,001	–	50	100.0
1.5B	valid	B	180	1	7,004	–	55	80.0
1.5B	valid	B	180	2	6,960	0	37	85.0
1.5B	valid	B	540	0	21,084	–	50	100.0
1.5B	valid	C	180	0	7,092	–	95	54.2
1.5B	valid	C	180	1	7,090	–	1	50.8
1.5B	valid	A-full	1600	0	3,200	–	50	100.0
1.5B	valid[cot]	base	0	0	0	5	56	66.7
1.5B	valid[fewshot4]	base	0	0	0	0	30	68.8
1.5B	div2	base	0	0	0	0	47	93.8
1.5B	div2	A	180	0	360	0	50	100.0
1.5B	div2	B	180	0	10,800	0	50	100.0
1.5B	div3	base	0	0	0	0	29	78.3
1.5B	div3	A	180	0	360	0	0	50.0
1.5B	div3	A	180	1	360	0	77	62.5
1.5B	div3	A	180	2	360	0	15	49.6
1.5B	div3	B	180	0	10,800	0	50	100.0
1.5B	div3	B	180	1	10,800	0	50	100.0
1.5B	div3	B	180	2	10,800	0	50	100.0
1.5B	div11	base	0	0	0	32	14	55.0
1.5B	div11	A	180	0	360	0	9	50.8
1.5B	div11	A	180	1	360	0	7	49.2
1.5B	div11	A	180	2	360	0	9	50.4
1.5B	div11	B	180	0	10,800	0	50	99.2
1.5B	div11	B	180	1	10,800	0	49	99.2
1.5B	div11	B	180	2	10,800	0	50	100.0
1.5B	div13	base	0	0	0	2	23	50.4
1.5B	div13	A	180	0	360	0	0	49.6
1.5B	div13	A	180	1	360	0	58	50.4
1.5B	div13	A	180	2	360	0	8	51.2
1.5B	div13	A	360	0	720	0	0	50.0
1.5B	div13	B	180	0	10,800	0	76	54.6
1.5B	div13	B	180	1	10,800	0	18	49.2
1.5B	div13	B	180	2	10,800	0	49	57.9
1.5B	div13	B	360	0	21,600	0	27	75.4
1.5B	div13	B	540	0	32,400	0	46	94.6
1.5B	div13	C	180	0	10,824	0	49	45.0
1.5B	div7_6d	base	0	0	0	0	88	52.5
1.5B	div7_6d	A	180	0	360	0	7	47.9
1.5B	div7_6d	A	180	1	360	0	9	50.8
1.5B	div7_6d	B	180	0	14,040	0	48	96.2
1.5B	div7_6d	B	180	1	14,040	0	43	90.0
3B	prime	base	0	0	0	110	2	47.1
3B	prime	A	180	0	360	0	67	82.9
3B	prime	A	180	1	360	0	63	81.7
3B	prime	B	180	0	13,370	0	60	90.4
3B	prime	B	180	1	13,260	0	41	79.6
3B	prime	S	180	0	19,736	49	22	69.6
3B	prime→prime_hard	A	180	0	0	0	99	50.4
3B	prime→prime_hard	A	180	1	0	0	92	47.9
3B	prime→prime_hard	B	180	0	0	0	100	50.4
3B	prime→prime_hard	B	180	1	0	0	69	53.8
3B	div7	base	0	0	0	0	55	94.2
3B	div7	A	180	0	360	0	5	54.6
3B	div7	A	180	1	360	0	100	50.0
3B	div7	B	90	0	5,400	0	45	60.0
3B	div7	B	180	0	10,800	0	27	68.3
3B	div7	B	180	1	10,800	0	41	82.1

model	cell	arm	n	seed	train tokens	cap	Yes (%)	acc. (%)
3B	div7	B	270	0	16,200	0	48	97.9
3B	div7	B'	180	0	10,800	51	96	50.8
3B	div7	D	180	0	10,800	0	100	50.0
3B	div7	S	180	0	11,130	0	52	97.5
3B	div7	S	180	1	11,198	0	54	96.2
3B	div7→div7_6d	A	180	0	0	0	0	50.4
3B	div7→div7_6d	B	180	0	0	0	22	54.2
3B	div7→div7_6d	B	180	1	0	0	33	65.0
3B	valid	base	0	0	0	10	52	85.8
3B	valid	A	180	0	360	0	50	100.0
3B	valid	A	180	1	360	0	50	100.0
3B	valid	B	180	0	7,001	0	59	91.2
3B	valid	B	180	1	7,004	0	50	100.0
3B	valid	S	180	0	30,284	0	57	92.1
3B	div3	A	180	0	360	0	50	99.2
3B	div3	A	180	1	360	0	50	99.2
3B	div3	B	180	0	10,800	0	50	100.0
3B	div3	B	180	1	10,800	0	50	100.0
3B	div11	A	180	0	360	0	6	55.4
3B	div11	A	180	1	360	0	1	50.8
3B	div11	B	180	0	10,800	0	50	99.6
3B	div11	B	180	1	10,800	0	50	100.0
3B	div13	base	0	0	0	0	53	92.1
3B	div13	A	180	0	360	0	0	50.0
3B	div13	A	180	1	360	0	28	48.3
3B	div13	A	360	0	720	0	0	50.0
3B	div13	A	540	0	1,080	0	0	50.0
3B	div13	B	180	0	10,800	0	70	52.9
3B	div13	B	180	1	10,800	0	42	59.2
3B	div13	B	360	0	21,600	0	33	83.3
3B	div13	B	540	0	32,400	0	50	99.6
3B	div7_6d	A	180	0	360	0	1	49.2
3B	div7_6d	B	180	0	14,040	0	35	82.5
7B	prime	base	0	0	0	4	30	72.1
7B	prime	A	180	0	360	0	1	51.2
7B	prime	A	180	1	360	0	67	82.5
7B	prime	B	180	0	13,370	1	60	89.2
7B	prime	B	180	1	13,260	0	36	80.0
7B	prime	S	180	0	26,737	25	32	72.1
7B	div7	base	0	0	0	1	45	82.5
7B	div7	A	180	0	360	0	35	49.2
7B	div7	A	180	1	360	0	75	52.9
7B	div7	B	180	0	10,800	0	43	86.7
7B	div7	B	180	1	10,800	0	46	90.4
7B	div7	B'	180	0	10,800	3	16	53.3
7B	div7	D	180	0	10,800	0	39	47.9
7B	div7	S	180	0	10,803	0	41	82.1
7B	valid	base	0	0	0	3	42	89.6
7B	valid	A	180	0	360	0	50	100.0
7B	valid	B	180	0	7,001	0	92	58.3
7B	div11	A	180	0	360	0	26	55.4
7B	div11	B	180	0	10,800	0	49	98.8
7B	div13	base	0	0	0	0	47	80.0
7B	div13	A	180	0	360	0	25	45.4
7B	div13	A	180	1	360	0	6	47.9
7B	div13	B	180	0	10,800	0	26	61.3
7B	div13	B	180	1	10,800	0	35	65.8

Table 7: Preregistered cells in detail. Qwen2.5-1.5B-Instruct, 4 digits and $n = 180$ unless stated; accuracy in %; seed 0 unless stated. For S7 the B column is the per-step accuracy q (written p in the preregistration text); for S8 and S10 it is the accuracy of arms D and S. –: not run.

	cell	probe	A	B	criterion	met
S1	div7, s1 / s2	46.5	50.8 / 47.9	90.0 / 82.5	$B \geq 80, B-A \geq 20$ pp, $p < 0.01$ in each seed	passed
S2	div3	50.7	54.0	100.0	≥ 2 of the 3 with $B-A$	passed
	div11	50.3	50.1	99.4	≥ 15 pp in both seeds	
	div13	53.3	50.4	53.9		
S3	div2	100.0	100.0	100.0	$A \geq 95$	passed
S4	div7, 3B model		52.3	75.2	$B > A$	passed
S5	div7 (6 digits), s0 / s1		49.4	96.2 / 90.0	$ B - q^6 \leq 10$ pp; $q^6 = 75.8$	failed
	secondary: 4-digit B			77.1	same	
S6	prime $\rightarrow$ hard set		49.6	–	$A < 60$; agrees more with the rule than with the label	passed
S7	(a) div13, $n = 540$			0.980	$q \geq 0.95$, acc. ≥ 85 (94.6)	holds
	(b) div7, $n = 270$			0.997	$q \geq 0.97$	holds
	(c) div7, $n = 90$			0.736	$q \leq 0.80$, acc. ≤ 75 (62.1)	holds
	(d) div11			0.994	$0.566 < q < 0.955$	fails
	(e) div3 / div2			1.000 / 0.993	$q \geq 0.97$	holds
S8	div7, arm D, s0 / s1			50.0 / 50.4	$D \leq 60$ in both seeds	passed
S9	div13, $n = 360$		50.0	75.4	$B \geq 80, A \leq 60$	failed
S10	div7, arm S			77.9	$S \leq 65$ in seeds 0 and 1; valid $S \geq 90$	failed

Table 8: Earlier answer-only study (same model and LoRA configuration; 240 training items, answer-only prompts, 3–4-digit numbers for arithmetic; mean of two seeds). Correct labels vs. labels permuted across training items.

task	base	correct labels	shuffled labels
divisible by 3	55.0	67.3	49.6
divisible by 7	53.3	51.0	52.5
divisible by 13	54.2	50.0	50.0
perfect square	51.2	83.3	51.5
prime	50.0	82.9	50.0
argument validity	66.7	99.4	50.0
joint consistency (SAT)	50.0	99.8	48.5

B SETUP DETAILS

Base-model cells whose generations often hit the cap are lower bounds: 3B prime base 47.1% (110 of 240 capped), 1.5B prime with 4 demonstrations 39.2% (104 capped) and 1.5B div11 base 55.0% (32 capped). The 1.5B valid base cell (30.0%) has no saved generations and is unaudited. Arm A collapses to a constant answer in single seeds (Yes rate near 0 or 100% in Table 6): 1.5B div3 and div13 seed 0, 3B div7 seed 1 and div13 seed 0; at larger n , 1.5B div7 at $n = 540$, 1.5B div13 at $n = 360$ and 3B div13 at $n = 360$ and 540 (all “No”); 3B div7 $\rightarrow$ div7_6d (“No”) and 3B div7_6d (Yes rate 1%); and 7B prime seed 0, which answers “No” on 237 of 240 items. A single untuned configuration is used throughout, so these collapses are part of what answer-only training did here, not a tuned optimum.

C THE LOGIC TASK

Arguments instantiate 6 valid forms (modus ponens, modus tollens, hypothetical syllogism, disjunctive syllogism, constructive dilemma, contraposition) and 6 invalid ones (affirming the consequent, denying the antecedent, affirming a disjunct, a broken chain, the converse, and “or” to “and”) with distinct random atom letters, rendered in English (“if A then B ”, “not A ”, “ A or B ”). The schema

Table 9: Training and evaluation setup, shared by every arm and model.

models	Qwen2.5-0.5B/1.5B/3B/7B-Instruct, bf16; 0.5B and 1.5B on a laptop GPU, 3B and 7B on one RTX 5090
adapter	LoRA $r = 8$, $\alpha = 16$, dropout 0.05, on q/k/v/o projections only
optimization	AdamW, learning rate 2×10^{-4} , batch size 1, 3 epochs, loss on completion tokens only
prompt	the model’s chat template; one user turn ending “End your reply with ‘Answer: Yes’ or ‘Answer: No’.”
seed	selects the balanced training sample and seeds PyTorch (adapter initialization, data order)
decoding	greedy; generation cap 176 tokens (div2/3/7), 192 (div11/13), 240 (6-digit div7), 512 (prime), 320 (valid), 512 for the CoT prompt
extractor	last “Answer: Yes/No” in the completion, else the first standalone Yes/No; otherwise unparsed; unparsed and truncated answers count as wrong
pools	valid: 1,600 training arguments; prime: 1,320 numbers (arm A-full); test items never in a pool
arm D	for each training item, the correct trace of another pool number with the same label
arm S	$K = 4$ samples per item at temperature 0.7 from the base model’s CoT prompt; one accepted completion per item

proposes an argument and Z3 decides it: the argument is labeled valid iff the premises together with the negated conclusion are unsatisfiable. For a valid argument the trace names the unsatisfiable core; for an invalid one it gives the satisfying assignment as a countermodel. Training and test share the 12 forms and never an instance, so a model can reach high accuracy by recognizing the form, which the atom-renamed n -gram probe does (100.0%). The task therefore checks that answers work when the surface determines the label; it is not a test of logical reasoning. Its results: base 30.0%, A 99.2% (99.2 and 100.0), B 88.3% (100.0 and 80.0), C 52.5% (54.2 and 50.8); at $n = 60$, A 97.5% and B 87.5%; at $n = 540$, both 100.0%.

D TRACE FORMATS

All arms share the prompt, which ends “End your reply with ‘Answer: Yes’ or ‘Answer: No’.” Examples of training completions, verbatim:

div7, B (prompt: “Is 1617 divisible by 7?”):

```
Process the digits of 1617 keeping a running remainder mod
7.  r = (10*0 + 1) mod 7 = 1.  r = (10*1 + 6) mod 7 = 2.  r
= (10*2 + 1) mod 7 = 0.  r = (10*0 + 7) mod 7 = 0.  Final
remainder 0, so 1617 is divisible by 7.
Answer:  Yes
```

div7, C (a scrambled trace from another example; the answer line keeps this example’s label):

```
running mod r = so = 4067 Process by 7.  7 a = 0.  0) 0.  7
mod (10*5 = r 7 the = = keeping + r is mod mod 4) (10*0 + of
4067 r + 7) 5.  = + digits remainder 7.  (10*0 = 7 0,
divisible Final 6) 4.  (10*4 mod remainder
Answer:  Yes
```

prime, B, a composite (“Is 4143 a prime number?”):

```
Check prime divisors up to isqrt(4143) = 64.  4143 mod 2 =
1.  4143 mod 3 = 0.  3 divides 4143, so 4143 is composite.
Answer:  No
```

prime, B, a prime (“Is 1493 a prime number?”):

```
Check prime divisors up to isqrt(1493) = 38.  1493 mod 2 =
1.  1493 mod 3 = 2.  1493 mod 5 = 3.  1493 mod 7 = 2.  1493
```

mod 11 = 8. 1493 mod 13 = 11. 1493 mod 17 = 14. 1493 mod
 19 = 11. 1493 mod 23 = 21. 1493 mod 29 = 14. 1493 mod 31
 = 5. 1493 mod 37 = 13. No prime divisor up to 38, so 1493
 is prime.
 Answer: Yes

valid, B, invalid (“Premises: (1) if H then D; (2) D. Conclusion: H.”):

Assume every premise is true and the conclusion false. That
 is satisfiable: set D=true, H=false. Under this assignment
 all premises hold while the conclusion is false, a
 countermodel, so the argument is invalid.
 Answer: No

valid, B, valid (“Premises: (1) if S then R; (2) S. Conclusion: R.”):

Assume every premise is true and the conclusion false, i.e.
 $\sim(R)$ holds. Together these force a contradiction: $\{S \rightarrow R,$
 $S, \sim(R)\}$ cannot all hold at once. Since assuming the
 conclusion false is impossible, the conclusion follows.
 Answer: Yes

A (every task): Answer: Yes or Answer: No.

E PREREGISTRATION TEXT AND DECISION LOG

Provenance of the prior div7 result. A verdict written on 2026-08-04 from the 60-example trace run alone (50.0%) stated that traces do not unlock divisibility by 7, while the preregistered 180-example cell was still running; that cell finished at 92.5% with no code change. The dose curve shows why both were right about their own n . The replication of that cell was the go/no-go condition S1, and it passed (90.0% and 82.5% in seeds 1 and 2).

First preregistration (prior grid). Written before any run of the prior grid, according to the history of the repository in which it was written, and never edited afterwards; results were appended below it. The file enters the present repository with the first commit of the manifest (P1), so its earlier date rests on the original repository. It fixed: arms A, B, C, an answers-only arm at larger scale, arms using model-generated traces, and a conditional wrong-trace arm; Qwen2.5-1.5B-Instruct with the LoRA configuration of §3; tasks prime, valid and div7, all reported; 240 problem-disjoint test items per task (120 per class), identical across arms, graded by one parser on the final answer. Endpoints: H1, $B > A$ at $n = 180$ per task, paired McNemar, Holm across the three tasks; H2, $B > C$ at $n = 180$; H3, accuracy at matched training tokens; H4, on div7 B reaches about 60% with A at chance. The full text is in the supplementary material.

Second preregistration and its decision log. Reproduced from PREDICTIONS.md up to the S10 entry; later entries (the S10 outcome, a wording note, and S11, a second-model-family test none of whose cells had run at submission) are in the supplementary material. Markdown markup converted, commit hashes replaced by the anonymized identifiers P1–P6 of the timestamp manifest and the location in each time stamp removed, text otherwise unchanged. The file’s p (per-step accuracy) is q in the paper. The header’s time was first written as an estimate (“~20:35”) and later replaced by the commit time; the decision log notes this. After the runs, a dated note was appended recording that the seed-0 “re-run” was a deterministic re-execution.

Preregistered predictions and decision rules (frozen before any run listed here).
 Committed 2026-09-25 20:29:47 (commit P1), before any of the runs below were started. The commit hash of this file is the preregistration. Earlier results (results/THINKING_VS_DATA.md, frozen 2026-08-04, and its later staged results block) are prior data, reported as such.

Provenance of the prior div7 result (stated in the paper). The 2026-08-04 verdict “H4 (a procedure trace unlocks div7): NO” was written from the $n = 60$ trace run alone ($B@60 = 50.0\%$). The preregistered main cell, B@180 seed 0, finished afterwards at 92.5% with the same worker (no code change). The paper reports the premature verdict and the later result.

Claim under test. Exact labels teach shortcuts; exact traces teach short programs. With labels from an exact verifier: answer-only supervision (arm A) recovers roughly what a simple probe on the answer surface recovers; trace supervision (arm B: procedure trace, then answer) succeeds when the per-step trace accuracy p , compounded over k steps, stays high (predicted accuracy about p^k), and fails when the procedure is long.

Decision rules. **S1 (go/no-go, before any other new run):** div7 B@180 seeds 1 and 2 each reach $\geq 80\%$ accuracy, with $B - A \geq 20$ percentage points and McNemar $p < 0.01$ against A of the same seed. If S1 fails, the paper is not submitted in this form. **S2:** at least 2 of {div3, div11, div13} (4-digit numbers) give $B - A \geq 15$ pp with the same sign in both seeds. **S3:** div2 A $\geq 95\%$ (the answer is surface-learnable from the last digit). **S4:** the div7 sign ($B > A$) holds for Qwen2.5-3B-Instruct. **S5 (falsifiable length test):** on 6-digit div7, B's accuracy is within 10 pp of p^6 , where p is B's per-step accuracy measured on 4-digit div7 generations. **S6 (shortcut):** A on prime agrees more with the rule "odd and no factor ≤ 7 " than with the true label on hard negatives (odd composites with no factor ≤ 7), where A's accuracy drops below 60%.

Definitions: a "win" is a difference of ≥ 10 pp with both seeds agreeing in sign; "neither" is both arms at or below majority class + 8 pp. McNemar tests per seed, Holm correction across tasks within each hypothesis family, $\alpha = 0.05$. Every cell is reported, including those that go against the claim.

Arms. base (zero-shot), A (answers), B (trace then answer), C (length-matched scrambled trace), B' (answer then trace, div7 only), plus eval-only controls: zero-shot chain-of-thought prompt and 4-shot trace demonstrations on the base model. Model: Qwen2.5-1.5B-Instruct (laptop), Qwen2.5-3B-Instruct and others where the RTX 5090 allows. LoRA $r = 8$, $\alpha = 16$, 3 epochs, lr $2e-4$, bf16, greedy decoding, $n = 180$ training examples, 240 problem-disjoint test items.

Decision log.

2026-09-25 21:09:47 (commit P2) — **S1 PASS.** div7 B@180 seed 1 = 90.0% vs A 50.8% (+39.2 pp, McNemar $p = 2e-18$); seed 2 = 82.5% vs A 47.9% (+34.6 pp, $p = 6.8e-13$). Runs in results/thinking_vs_data/runs_L1.jsonl, produced by the unchanged worker experiments/exp_thinking_ft_worker.py. The paper proceeds; the remaining tests run on experiments/exp_worker_v2.py.

2026-09-25 21:26:10 (commit P3; the div7_6d B seed-0 generation file was written at 21:27:30) — **S5 clarification, recorded before either S5 cell finished** (div7_6d B seed 0 was mid-training; the 4-digit-adaptor transfer eval had not started). Primary S5 cell = arm B trained and tested on the 6-digit task (div7_6d), seeds 0 and 1, compared with p^6 where p = per-step accuracy of 4-digit div7 B generations (seed 0: $p = 0.955$, $p^6 = 0.758$). Secondary = the 4-digit B adapter evaluated on 6-digit inputs. Also reported, labelled post hoc: the fraction of fully correct 6-digit traces vs p^6 , and answer accuracy vs $p^6 + (1 - p^6) * g$, where g is the answer-correct rate when the trace is wrong (4-digit: $g = 0.561$), since a wrong trace still yields the right yes/no answer about half the time.

2026-09-25 21:29:52 (commit P4) — **S5 (primary) FAILED as preregistered:** 6-digit div7 B seed 0 = 96.2% vs $p^6 = 75.8\%$ (+20.4 pp > 10 pp tolerance). Its own per-step accuracy is 98.8% (vs 95.5% for the 4-digit model), so p is not a fixed property that transfers between training sets.

2026-09-25 21:29:52 (commit P4; div11 B seed 0 started after 21:30:58) — **S7 (new hypothesis, preregistered before any of its cells ran):** per-step trace accuracy p is set by the number of supervised transitions per entry of the (remainder, digit) table, $m = k * n / (10 * d)$ (k steps per trace, n training examples, $10d$ table entries). Observed so far: div7 $n=180$ $m=10.3 \rightarrow p=0.955$; div7_6d $n=180$ $m=15.4 \rightarrow p=0.988$; div13 $n=180$ $m=5.5 \rightarrow p=0.566$. Predictions: (a) div13 B $n=540$ seed 0 ($m=16.6$): $p \geq 0.95$ and answer accuracy $\geq 85\%$; (b) div7 B $n=270$ seed 0 (4-digit, $m=15.4$): $p \geq 0.97$; (c) div7 B $n=90$ seed 0 ($m=5.1$): $p \leq 0.80$ and answer accuracy $\leq 75\%$; (d) div11 B $n=180$ ($m=6.5$, not yet run): $0.566 < p < 0.955$; (e) div3 B $n=180$ ($m=24$) and div2 B $n=180$ ($m=36$): $p \geq 0.97$. S7 passes if at least 4 of (a)-(e) hold; every cell is reported either way.

Note (21:35): the clock times in the entries above were first written as estimates; they have been replaced by the commit timestamps, which are the authoritative record.

2026-09-25 21:38:05 (commit P5) **S8 and S9, preregistered before their cells ran** (div11 B seed 0 = 99.2% had finished; none of the cells below had started): S8 (grounding control): div7 arm D (a fluent, correct long-division trace of a DIFFERENT number with the same label) at $n = 180$, seeds 0 and 1: accuracy $\leq 60\%$ in both seeds, i.e. within 10 pp of A, far below B. S9 (dose, step difficulty fixed): div13 B at $n = 360$, seed 0 ($m = 11.1$ transitions per table entry, close to div7's 10.3 at $n = 180$): accuracy $\geq 80\%$; div13 A at $n = 360$ seed 0 stays $\leq 60\%$.

2026-09-25 21:46:44 (commit P6) **S10, preregistered before any arm-S cell ran.** Arm S = self-generated traces filtered by the exact verifier on the final answer only (STaR / rejection-sampling fine-tuning, K = 4 samples per item at temperature 0.7 from the base model’s chain-of-thought prompt; one kept completion per item; same 180 items and LoRA as A/B). Prediction: outcome-only verification does not supply correct steps, so on div7 arm S does not unlock the task: div7 S accuracy $\leq 65\%$ in seeds 0 and 1, and fewer than half of the kept div7 traces have all remainders correct. On valid, where answers are surface-learnable, S $\geq 90\%$. Prime S is reported without a prediction.

F THE COMPOUNDING APPROXIMATION (POST HOC)

This derivation is post hoc: it was not part of the preregistered S5 rule, and it does not change S5’s verdict. Suppose each emitted state is correct with probability q , independently, and that after the first error the emitted final state is uniform on $\mathbb{Z}_d$. With probability q^k the trace is error-free and the answer is right. Otherwise the answer is right with probability $1/d$ on a multiple of d and $(d-1)/d$ otherwise, so on a balanced test set the accuracy is $q^k + \frac{1}{2}(1-q^k)[\frac{1}{d} + \frac{d-1}{d}] = (1+q^k)/2$. Both assumptions can fail: errors are correlated across steps (a model that miscomputes one table entry does so whenever the entry recurs), and a wrong state need not be uniform (on div13, wrong traces end at $r = 0$ in 179 of 235 cases). The positive-class accuracy, $q^k + (1-q^k)/d$, is the cleaner check, because a model that answers “no” by default scores well on non-multiples: on 6-digit div7 it is 88.8%.